

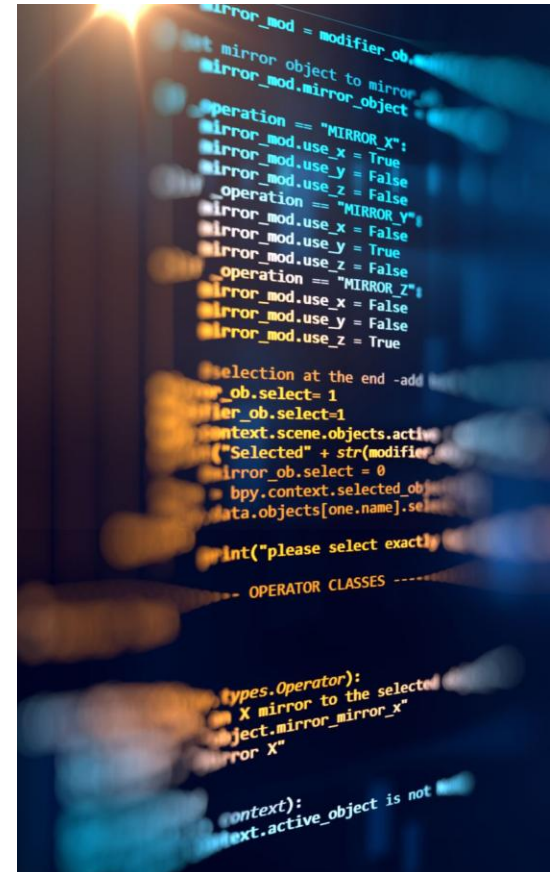
Introduction to Text Mining

By:-

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Definition of Text Mining

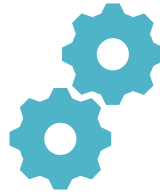
- Text mining is a knowledge-intensive process for extracting useful information from unstructured textual data.
- It combines techniques from Natural Language Processing (NLP), machine learning, and data mining to analyze textual information.
- The goal is to transform raw text into structured data for tasks such as classification, clustering, sentiment analysis, and trend detection.
- Text mining helps organizations uncover hidden information, support decision-making, and automate understanding of vast text sources.



Data Mining versus Text Mining



Both seek for novel
and useful patterns



Both are semi-
automated processes



Difference is the
nature of the data

Data Mining versus Text Mining



Structured versus
unstructured data

Structured data: in databases

Unstructured data: Word documents, PDF files, text excerpts, XML files, and so on

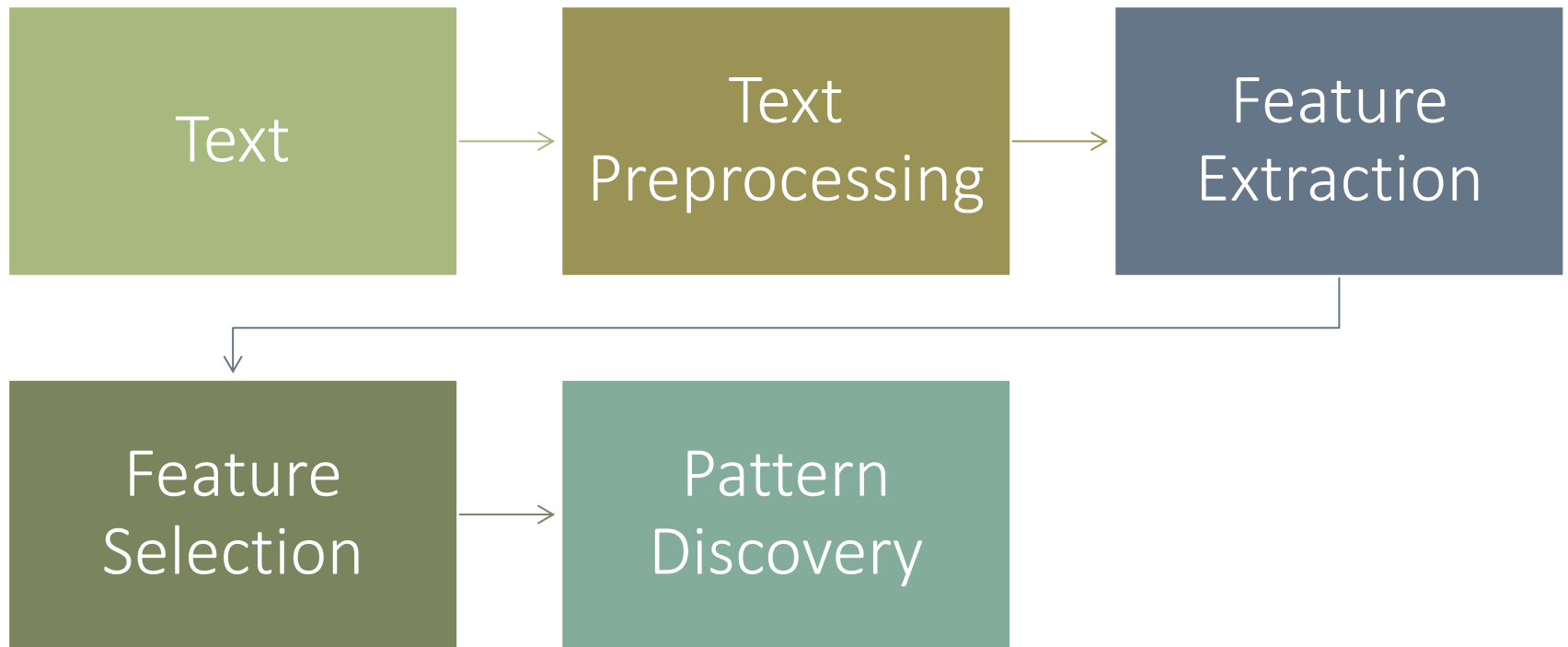


To perform text mining – first, impose structure to the data, then mine the structured data.

Why Text Mining?

- Roughly 85-90 percent of all corporate data is in some kind of unstructured form (e.g., text)
- Unstructured corporate data is doubling in size every 18 months...
- Tapping into these information sources is not an option, but a need to stay competitive
- A semi-automated process of extracting information and discovering knowledge from unstructured data source

Text Mining Process



Text Preprocessing

- Transforms unstructured data into a structured format suitable for analysis.
- include all those routines, processes, and methods required to prepare data for a text mining system's core knowledge discovery operations.
- Includes routines to prepare data for core knowledge discovery operations.

Feature Extraction



Extracts key elements like

Words

terms

Concepts



In order to represent
documents.

Feature Selection



Pick important features



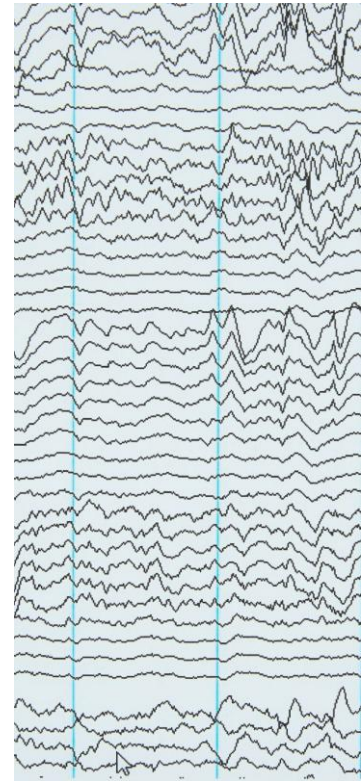
mainly used in connection with applying known machine learning and statistical methods on text when addressing tasks such as Document Clustering or Document Classification

Pattern Discovery

- Analyzes concept co-occurrence patterns across documents to discover relationships.

Ex:-

- *Classification*
- *Clustering*
- *Trend Analysis*
- *Anomaly Detection*



Use of Text Mining

- Benefits of text mining are obvious especially in text-rich data environments such as
 - *law (court orders), academic research (research articles), finance (quarterly reports),*
 - *medicine (discharge summaries), biology (molecular interactions), technology (patent files), marketing (customer comments), etc.*



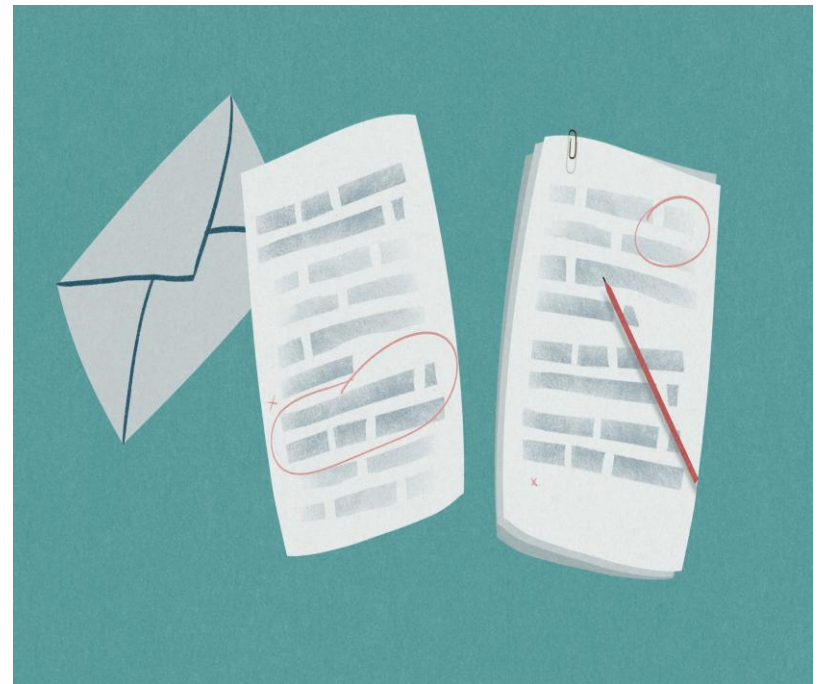
Use of Text Mining

- Electronic communication records (e.g., Email)
 - Spam filtering
 - Email prioritization and categorization
 - Automatic response generation



Today's Example

- Classification of free-text documents is a common task in the field of text mining.
- It is used to categorize documents
 - i.e. assign pre-defined topics,
 - or it can be used for sentiment analysis.



Today's Example

- Today we want to construct a workflow that
 1. *reads and preprocesses text documents,*
 2. *transforms them into a numerical representation*
 3. *builds a predictive model to assign pre-defined labels to documents.*



Today's Example

- Additional tasks:

- Sentiment analysis

- Visualization of documents

- Document clustering

- Topic Modeling

- Text Summarization



Today's Example

The image shows a screenshot of a TripAdvisor review for the restaurant "Burgermeister" in Berlin. The review is titled "Great late night burger joint" and is rated 4.5 stars. The reviewer is "shirlulit" from Heidelberg, Germany. The review text describes a late-night dining experience with great hamburgers, atmosphere, and music. The review is helpful, with 2 helpful votes. The screenshot also shows the restaurant's location, address, and a list of reviews sorted by date.

Rating

Title

Author

Full text

Was this review helpful? Yes

Problem with this review?

180 reviews from our community

Rating	Count
Excellent	112
Very good	47
Average	16
Poor	4
Terrible	1

180 reviews sorted by Date **Rating** **English first**

Great late night burger joint"

4.5 Reviewed March 3, 2013 **NEW**

I just had a burger there at 3 AM last night, They have great hamburgers, and awesome atmosphere and music. During the day people sit outside of the little burger 'shack' with smiles on their faces. When I got there there were no chairs, it was just people standing inside and eating (i guess due to weather) I liked it...

Today's Example

- Build a classifier to distinguish between reviews about Italian or Chinese restaurants.



Carmen D
Leamington Spa, United Kingdom

Reviewer

☆ 5 reviews

🍴 3 restaurant reviews

🌐 Reviews in 3 cities

👍 1 helpful vote

"So Italian"

★★★★★ Reviewed January 3, 2014

Excellent pizzas and pasta - very true to Italian style pizza places. Fast service, great inexpensive wine from a barrel, and excellent location to boot! They also do take away so you can eat it right on the canal. Come here very often, can't get enough.

Visited August 2013

Was this review helpful?

[Ask Carmen D about Il Casolare](#)

This review is the subjective opinion of a TripAdvisor member and not of TripAdvisor LLC.

Review about
an Italian or a
Chinese
restaurant?

Examples of Text Mining Applications

Product review sentiment analysis

Automatic email routing

Detecting fake news

Market trend monitoring

Customer service automation

Medical text analysis in healthcare

Social media monitoring

Risk & fraud detection



Common Challenges

- Data noise (slang, typos, emojis)
- Ambiguity in language
- Domain-specific terms
- Multilingual and code-mixed data
- High computational requirements
- Large dataset processing

Conclusion



Text mining systems transform unstructured data into structured formats



enabling advanced analysis and discovery of patterns and trends.

References

Feldman, Ronen, and James Sanger. *The text mining handbook: advanced approaches in analyzing unstructured data*. Cambridge university press, 2007.